

Xinai Zhang

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EDUCATION

Stanford University

M.S. in Mechanical Engineering

Stanford, CA

Expected May 2028

Brown University

Sc.B. (Honors) in Biomedical Engineering — GPA: 4.0/4.0

Providence, RI

Sep 2022 - May 2026

Tau Beta Pi Engineering Honor Society Member — Top 1/8 of Junior Class and Top 1/5 of Senior Class

WORK EXPERIENCE

Cornerstone Robotics

Clinical Development Engineer Intern

Jun 2025 – Aug 2025

Hong Kong

- Conducted clinical development research on curved bipolar dissectors; analyzed 230+ adverse event reports in the MAUDE database and identified 8+ design-specific risk factors; produced a comprehensive risk analysis report to guide product iteration and inform downstream clinical strategy
- Supported 20+ porcine, canine, and cadaveric robotic-assisted surgical studies as circulating nurse and intraoperative assistant, managing pre-operative preparation and intraoperative coordination; documented pre-surgical clinical presentations and product performance issues, validated preclinical trial outcomes for the latest robot model
- Led development of 3 regulatory SOPs; built a streamlined project management and financial reporting system, and oversaw clinical documentation organization and progress tracking across concurrent workstreams

Coulombe Lab, Brown University School of Engineering

Undergraduate Researcher

Sep 2025 – May 2026

Providence, RI

- Developed a pulsatile stretch tissue-conditioning bioreactor system that conducted up to 40% stretch on engineered tissue and enhanced the functional performance of engineered bioelectric grafts for post-myocardial infarction cardiac repair
- Developed full mechanical design and CAD models, and wrote Arduino C++ firmware with TMC2209 microstepping control and UART realtime communication for highprecision linear actuation
- Calibrated displacement and velocity control via motion capture ($R^2 = 0.9924$), enabled 4 programmable stretch waveforms (uniform, asymmetric, trapezoidal, incremental) and validated system stability (amplitude CV<1% in 48 hrs)

Srivastava Lab, Brown University School of Engineering

Undergraduate Researcher

Jun 2024 – Sep 2024

Providence, RI

- Performed cell culture maintenance, passaging, and migration assays to investigate cellular response to mechanical stress
- Processed 200+ microscopy images and quantified stiffness-dependent cell damage and recovery using ImageJ, demonstrating accelerated scratch closure on rigid (100 kPa) gels vs. delayed repair on softer (1 kPa) substrates

PROJECT EXPERIENCE

Redesigning NASA Space Suit through Biomechanical Interventions

Undergraduate Researcher

Jan 2024 – May 2024

Providence, RI

- Modeled astronaut gait in OpenSim to design a passive elastic assist for space suits, addressing fatigue and injury issues
- Developed biomechanical testing protocols to quantify assistive torque and improve suit mobility
- Built and tested a neoprene-elastomer wearable prototype, validated up to 8 Nm of torque assistance
- Presented findings at NASA Rhode Island Space Grant Consortium

Muscle-Controlled Mario Kart with Electromyography Sensors

Undergraduate Researcher

Sep 2024 – Jan 2025

Providence, RI

- Developed a multi-stage EMG signal amplifier that converts muscle activity into control input for a Mario Kart emulator
- Developed signal-processing logic in Arduino (C++); calibrated discrimination threshold in LabVIEW for precise control

Brown Space Engineering

Structures and Avionics Group Member

Aug 2023 – Jan 2024

Providence, RI

- Developed CAD models for PCB access port components in Autodesk Fusion 360; validated multi-component compatibility in space-constrained configurations via 3D-printed prototypes
- Designed power system PCB in Autodesk EAGLE, enabling reliable solar-to-battery charge/discharge regulation

PUBLICATIONS

ACS Applied Materials & Interfaces

Co-Author

2022

DOI: 10.1021/acsami.1c25102

Li, S., Liu, D., Hu, Y., Su, Z., Zhang, X., Guo, R., Li, D., Lu, Y. (2022). Soft magnetic microrobot doped with porous silica for stability-enhanced multimodal locomotion in a nonideal environment. *ACS Applied Materials & Interfaces*, 14(8), 10856-10874

SKILLS

Engineering Software: Fusion 360, SolidWorks, MATLAB, LabVIEW, ImageJ, Arduino, OpenSim, Python, GraphPad

Laboratory: Cell Culture, Cell Assays, Animal Models, Western Blotting, Genotyping, Mechanical Testing, Soldering

Digital Media & Design: DaVinci Resolve, Adobe Premiere Pro, Final Cut Pro, Adobe Photoshop, Adobe InDesign